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First-line chemotherapy with tislelizumab for patients with extensive-stage small cell lung cancer: a cost-effectiveness analysis

Rong Long[1] & Fangping Chen[2,3]✉

The Phase 3 RATIONALE-312 trial (NCT04005716) showed that tislelizumab plus chemotherapy led to a noteworthy enhancement resulted in a significant improvement in overall survival among patients diagnosed with extensive-stage small-cell lung cancer (ES-SCLC) compared to chemotherapy alone. The treatment also had an acceptable level of safety. Nevertheless, the debate over the efficacy of implementing several treatment plans in competition continues due to the significant expenses involved. Therefore, we aimed to evaluate the potential efficacy and cost of tislelizumab treatment as a first-line treatment for the ES-SCLC patient population in China. The study assessed primary health outcomes by measuring life years (LYs), quality-adjusted life years (QALYs), and incremental cost-effectiveness ratios (ICERs). This was done using a Markov model considering three health states with a 15-year horizon. To assess its model resilience, we conducted one-way sensitivity analyses with probability. In addition, subgroup analyses of some pre-specified patients was performed. Compared to chemotherapy alone, tislelizumab plus chemotherapy resulted in an additional 0.34 ($8,028) QALYs, leading to an ICER of $23,553 per QALY for the overall patient population. The ICER was lower than the assumed willingness-to-pay threshold of $35,367 per QALY. Approximately 60% of simulations suggested that tislelizumab in combination with chemotherapy was cost-effective, while 40% suggested that chemotherapy alone was cost-effective. The subsequent sensitivity analyses revealed that the health utility value associated with the disease progression parameter had the greatest influence on ICER. Tislelizumab plus chemotherapy was a preferable treatment option for regimens for patients with ES-SCLC in China. This finding is important in guiding the Chinese healthcare system.

Keywords Extensive-stage small-cell lung cancer (ES-SCLC), Markov model, Cost-effectiveness, Quality-Adjusted life years (QALYs), Tislelizumab, First-line

Lung cancer is a prevalent form of cancer in China, causing a significant number of deaths. In 2021, the disease was responsible for about 870,000 fatalities[1,2]. Small cell lung cancer (SCLC) accounts for around 15% of lung cancer cases characterized by rapid growth, high aggressiveness, and a high-grade neuroendocrine carcinoma, with an extremely high mortality rate[3]. Small cell lung cancer has a high susceptibility to early metastasis, with most patients being detected at an advanced stage known as extensive-stage small cell lung cancer (ES-SCLC). The median overall survival (OS) typically ranges from about 10 months. Even in cases of early-stage illness, the OS of patients rarely surpasses 2 years[4]. Since the late twentieth century, etoposide-platinum chemotherapy has been the primary choice for first-line treatment of ES-SCLC because of its remarkable overall response rate (ORR) of ≥ 50%. However, almost all patients will quickly develop drug resistance, leading to a bleak prognosis.

Immune checkpoint inhibitors (ICIs) have recently been introduced and have demonstrated encouraging clinical efficacy in ES-SCLC treatment, presenting an exciting possibility and renewed optimism for ES-SCLC patients[5]. The activation of T-cells and the resulting immune-mediated destruction of tumors can be

[1]Department of Oncology, Xiangya Hospital, Central South University, Changsha 410008, Hunan, China. [2]Department of Stomatology, Changsha Stomatological Hospital, Changsha 410004, Hunan, China. [3]Department of Stomatology, School of Stomatology, Hunan University of Chinese Medicine, Changsha 410208, Hunan, China. ✉email: cabbagechen204@163.com

significantly enhanced when chemotherapy is provided simultaneously with ICI, leading to a synergistic effect[6]. The IMpower133 III study provided the foundation for the official approval of the indication for the PD-L1 inhibitor atezolizumab combined with etoposide/platinum as an initial treatment of ES-SCLC by China's National Medical Products Administration (NMPA) in February 2020, marking a new era in immunotherapy. Subsequently, introducing ICIs to first-line chemotherapy in ES-SCLC patients has improved survival in several phase 3 randomized trials that have become the new standard care for first-line treatment in ES-SCLC[7]. However, the autonomy to select a therapeutic course of action about efficacy, safety, and feasibility in clinical environments remains limited.

In a recent phase 3 study (NCT04005716), patients with untreated ES-SCLC were treated with tislelizumab, a monoclonal antibody that selectively binds to the CC' loop of PD-1 and functionally optimizes specific segments to achieve slow-dissociated blockage of PD-L1[8], which exhibited a survival benefit that was both statistically significant and clinically meaningful while maintaining a manageable safety profile[9]. Patients enrolled in this phase 3, multicenter, randomized, double-blind clinical trial experienced a significant improvement in the OS when tislelizumab was administered in conjunction with chemotherapy, compared to chemotherapy alone. The OS was increased from 15.5 months to 13.5 months, with a hazard ratio (HR) of 0.75 (95% confidence interval [CI] 0.61–0.93, P = 0.0035). The progression-free survival (PFS) also showed improvements, with 4.8 months compared to 4.3 months and an HR of 0.64 (95% CI 0.52 to 0.78, P = 0.0001)[9]. Consequently, the NMPA has authorized tislelizumab for a new indication, supporting its combined use with etoposide and platinum-based chemotherapy for the first-line treatment of patients with ES-SCLC.

While tislelizumab presents a promising and innovative immunotherapy choice for patients receiving initial treatment for ES-SCLC, we must consider its substantial cost alongside its enhanced therapeutic advantages. This is crucial to ensure its affordability for average patients and to prevent excessive strain on the healthcare system.

Materials and methods

Patient population and intervention

Based on basic clinical data from the phase 3 RATIONALE-312 trial, participants were included with confirmed ES-SCLC who were randomly assigned to receive either tislelizumab (n = 227) or placebo (n = 229) in addition to chemotherapy. Both groups of patients were given either tislelizumab or a placebo intravenously every three weeks at a dose of 200 mg, along with chemotherapy (carboplatin (AUC 5) or cisplatin (75 mg/m^2), D1 + etoposide (100 mg/m^2), D1-3 for four cycles). After that, patients continued to receive tislelizumab or placebo (200 mg) every 3 weeks until their disease progressed or they experienced intolerable side effects. At that point, second-line therapy was administered to patients in both tislelizumab plus chemotherapy (TC) and chemotherapy (C) groups. All patients underwent imaging every six weeks to detect progressive disease (PD). After diagnosing PD, 59.9% of patients in the TC group and 73.9% in the C group patients received topotecan (1.25 mg/m^2/d administered intravenously on D1-5 every three weeks), following the latest study guidelines[10,11]. In this study, the Chinese participants included were assumed to have specific characteristics for calculating therapeutic dosing regimens[12,13]. The patients who did not receive further treatment were presumed to have the best supportive care (BSC) until their death, along with palliative care for those who died due to treatment-related causes. Table S2 details administration methods, drug dosages, and unit prices.

Our investigation was based on previously conducted studies and did not contain any new studies with humans and/or use human or animal tissue or samples performed by any of the authors, conducted following the established guidelines for reporting health economic evaluations, known as the standard health economic evaluation reporting standards (CHEERS)[14], as depicted in Table S3.

Model design

TreeAge Pro 2022 software was used to mimic the disease development of individuals with ES-SCLC. The model included three distinct health states: progression-free survival (PFS), PD, and death. At the beginning of the treatment, each patient was in a condition called PFS. They had the choice to either stay in the PFS state or move to either the (PD state or the state of death at each cycle. This decision depended on the transition probabilities, as presented in Fig. S1.

As precise baseline patient data from the RATIONALE-312 trial were unavailable, GetData Graph Digitizer (v 2.26) was used for the analysis[15]. As per the Akaike and Bayesian information criterion, it was determined that the Weibull distribution was the optimal choice for precisely rebuilding the survival curves of both the TC and C groups upon visual analysis (Fig. S2 and Table S2)[16]. The distributions of the γ (scale) and λ (shape) parameters were later analyzed using Rstudio (v 1.2.5042) (Table 1).

Within this state-transition model, each cycle had a fixed duration of 6 weeks, and the overall model covered 15 years. It was projected that 99% of patients would have passed away throughout this period. The key factors considered in this analysis were the overall cost, quality-adjusted life years (QALYs), life years (LYs), incremental cost-effectiveness ratio (ICER), and incremental net health benefit (INHB). These factors were evaluated based on the willingness-to-pay (WTP) thresholds set at $35,367 per QALY, three times the Chinese GDP in 2022, per the guidelines provided by the World Health Organization (WHO). Furthermore, the total costs and health values were annually discounted at 5% by the recommendations of the Chinese Pharmacoeconomic Evaluation Guide for the year 2020[17].

Expenditure input and utility

The direct medical expenses included drug costs, the management of treatment emergent adverse events (TEAEs), expenses for laboratory examinations, radiological tests, administration, best supportive care (BSC), and palliative care, as depicted in Table S1. The costs associated with the intervention were obtained from relevant

Variable	Mean value (Range)	Reference	Distribution
Clinical parameters			
Weibull survival model for tislelizumab plus chemotherapy			
OS	Scale = 0.035185, Shape = 1.045204	–	–
PFS	Scale = 0.27884, Shape = 0.61923	–	–
Weibull survival model for chemotherapy			
OS	Scale = 0.01429, Shape = 1.457591	–	–
PFS	Scale = 0.25393, Shape = 0.65114	–	–
Rate of post-discontinuation therapy			
Tislelizumab plus chemotherapy group	0.599 (0.4792–0.7188)	9	Beta
Chemotherapy group	0.739 (0.5912–0.8868)	9	Beta
Risk for most common TEAEs in tislelizumab plus chemotherapy group			
Anemia	0.846 (0.6768–1.0152)	9	Beta
Alopecia	0.789 (0.6312–0.9468)	9	Beta
Nausea	0.432 (0.3456–0.5184)	9	Beta
Neutropenia	0.687 (0.5496–0.8244)	9	Beta
Thrombocytopenia	0.498 (0.3984–0.5796)	9	Beta
White blood cell count decreased	0.559 (0.4472–0.6708)	9	Beta
Risk for most common TEAEs in chemotherapy group			
Anemia	0.847 (0.6776–1.0164)	9	Beta
Alopecia	0.790 (0.632–0.948)	9	Beta
Nausea	0.432 (0.3456–0.5184)	9	Beta
Neutropenia	0.703 (0.5624–0.8436)	9	Beta
Thrombocytopenia	0.520 (0.416–0.624)	9	Beta
White blood cell count decreased	0.646 (0.5168–0.7752)	9	Beta
Utility and disutility			
Utility of PFS	0.673 (0.5384–0.8076)	22	Beta
Utility of PD	0.473 (0.3784–0.5676)	22	Beta
Disutility of Anemia	0.073 (0.0584–0.0876)	23	Beta
Disutility of Alopecia	0.058 (0.0464–0.0696)	23	Beta
Disutility of Nausea	0.125 (0.100–0.150)	23	Beta
Disutility of Neutropenia	0.20 (0.16–0.24)	23	Beta
Disutility of Thrombocytopenia	0.190 (0.152–0.228)	23	Beta
Disutility of White blood cell count decreased	0.20 (0.16–0.24)	23	Beta
Cost, $/per cycle			
Treatment cost			
Tislelizumab	758 (606.4–909.6)	Local Hospital	Gamma
Etoposide	9 (7.2–10.8)	Local Hospital	Gamma
Cisplatin	28 (22.4–33.6)	Local Hospital	Gamma
Carboplatin	108 (86.4–129.6)	Local Hospital	Gamma
Topotecan	298 (238.4–357.6)	Local Hospital	Gamma
Cost of TEAEs			
Tislelizumab plus chemotherapy group	1282 (1025.6–1538.4)	17,18	Gamma
Chemotherapy group	1347 (1077.6–1616.4)	17,18	Gamma
Laboratory	154 (123.2–184.8)	12	Gamma
Radiological test	471 (376.8–565.2)	12	Gamma
Adminstration	33 (26.4–39.6)	12	Gamma
Best supportive care	205 (164–246)	12	Gamma
Palliative care per patient	2065 (1652–2478)	12	Gamma
Discount rate	0.05 (0–0.08)	16	Uniform

Table 1. Key Parameters. *OS* overall survival, *PFS* progression-free survival, *TEAEs* treatment-emergent adverse events, *PD* progressive disease.

local hospital data, whereas other direct medical expenses were derived from previous reports[12,18,19]. When calculating the dose of the intervention drugs, based on the trial, we assumed that an average Chinese patient was 61 years old, 1.64 m tall and weighs 65 kg, with a corresponding body surface area (BSA) of 1.72 square meters, a serum creatinine level of 1 mg/dL, and an area under the curve (AUC) of 5 mg/mL/min[9,20,21]. Notably,

the analysis did not consider mild adverse effects in grades 1 and 2 and only focused on the most frequent TEAEs with disutility values. The follow-up costs included charges for radiological tests conducted every six weeks during the treatment cycles. Due to government regulation, drug prices in China remain unadjusted for inflation to ensure stability in healthcare expenses. According to the exchange rate of $1 = ¥7.2695, all expenses are stated in US dollars.

The quality of life (QoL) of a patient was described using utility values, ranging from 0 (death) to 1 (ideal health) as an illness progressed naturally. Unfortunately, the RATIONALE-312 trial did not provide comprehensive patient utility values. Thus, previous researchers specifically examined ES-SCLC with utility values[22]. The disutility values associated with adverse events (AEs) were obtained from published literature[23]. The cost data in this investigation conformed to the gamma distribution, while the utility value and the incidence of adverse events followed the beta distribution[24].

Sensitivity analyses

A one-way sensitivity analysis evaluated the model's stability and uncertainty. In this analysis, important parameters were randomly changed within a range of ± 20% of their baseline values to find the ones that significantly affected the model's results[25,26]. To simulate the effect of changing several uncertain parameters simultaneously on the study results, a 10,000-iteration Monte Carlo sampling was used to perform a probability sensitivity analysis[25]. The present analyses were represented using tornado diagrams and visualized using cost-effectiveness acceptability curves and scatter plots. In addition, subgroup analyses of some pre-specified patients were performed using the method of Hoyle et al.[27].

Results

Base-case analysis

Utilizing a well-established model spanning 15 years, Chinese patients with ES-SCLC who received first-line tislelizumab along with chemotherapy experienced a significant increase in life expectancy of 0.70 LYs (8.40 months) compared to those who were given placebo in addition to chemotherapy. Upon considering modifications in QoL, the incremental cost (attributed to improved efficacy) of tislelizumab plus chemotherapy in China amounted to $8,028 (0.34 QALYs) compared to placebo combined with chemotherapy. Accordingly, the corresponding ICER (INHB) values were $23,553/QALY (0.11 QALYs) at respective WTP thresholds of $35,367/QALY (Table 2).

Sensitivity analysis

Figure 1 depicts examined parameters significantly influencing the ICER value. The outcomes of the analysis indicated that variations within specified ranges did not change the results analyzed significantly. The health utility value of the PD had the greatest influence on the ICER, which ranged from 0.5676 to 0.3784, leading to an ICER from $19,877 to $28,896 per QALY. Moreover, the cost of tislelizumab and the risk of adverse events in both TC and C groups moderately impacted the ICER. Conversely, other variables, such as the utility value of PFS, BSC cost, and AEs disutility values, demonstrated little or negligible influence on the model.

Our probabilistic sensitivity analysis indicates that tislelizumab combined with chemotherapy is cost-effective for the entire Chinese patient population. This cost-effective combination regimen becomes more likely when the WTP threshold rises, as shown by the cost-effectiveness acceptability curve (Fig. 2). The scatter plot (Fig. S3) indicates the probability of tislelizumab combined with chemotherapy being economically advantageous based on 10,000 Monte Carlo simulations, wherein 60% of simulations fall below the WTP line.

We also conducted cohort analyses for specific subgroups, such as sex and smoking status (Table 3). In the female, never-smoking, stage III patient subgroups, tislelizumab plus chemotherapy yielded an additional 0.24 QALYs (0.50 LYs), 0.23 QALYs (0.48 LYs), and 0.31 QALYs (0.65 LYs) compared to chemotherapy alone, resulting in ICERs of $33,329, $34,138, and $28,643 per QALY, with INHB values of 0.01, 0.01, and 0.06 QALYs, respectively, each with a cost-effectiveness probability exceeding 50%. For patients with distant metastases, particularly those with liver metastases at baseline, tislelizumab plus chemotherapy demonstrated higher cost-effectiveness, providing an additional 0.28 QALYs, resulting in an ICER of $30,469 per QALY and an INHB of 0.04 QALYs. In contrast, a significant portion of the subgroup patients were less cost-effective.

Discussion

The IMpower133 and CASPIAN studies conducted in 2018 confirmed that immunotherapy combined with chemotherapy considerably improved the prognosis of ES-SCLC. This treatment approach resulted in a median OS of 12 to 13 months, setting a new standard for the initial treatment of ES-SCLC[28,29]. The "Chinese Society of Clinical Oncology (CSCO) SCLC Diagnosis and Treatment Guidelines" quickly endorsed it as a level I recommendation for initial therapy of ES-SCLC patients, marking a significant advancement in SCLC treatment[30].

Treatment	Total cost, $	Overall LYs	Overall QALYs	ICER, $		INHB, QALY
				per LY	per QALY	WTP($35,367/QALY)
Chemotherapy	16,080	1.57	0.68	Reference	Reference	Reference
Tislelizumab plus chemotherapy	24,107	2.28	1.02	11,422	23,553	0.11

Table 2. Results of the Base-Case Analysis. *LYs* life-years, *QALYs* quality-adjusted life-years, *ICER* incremental cost-effectiveness ratio, *INHB* incremental net health benefits, *WTP* Willingness-to-pay.

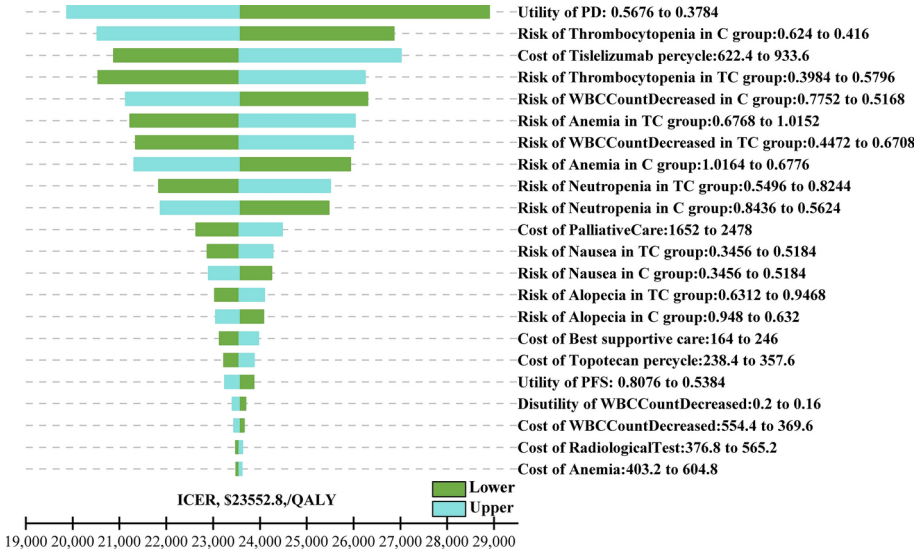

Fig. 1. The one-way sensitivity analyses for tislelizumab plus chemotherapy strategy compared to chemotherapy strategy. *PD* progressive disease, *TC* tislelizumab plus chemotherapy, *C* chemotherapy, *WBC* White blood cell, *PFS* progression-free survival, *WTP* willingness-to-pay, *ICER* incremental cost-effectiveness ratio, *QALY* quality-adjusted life-year.

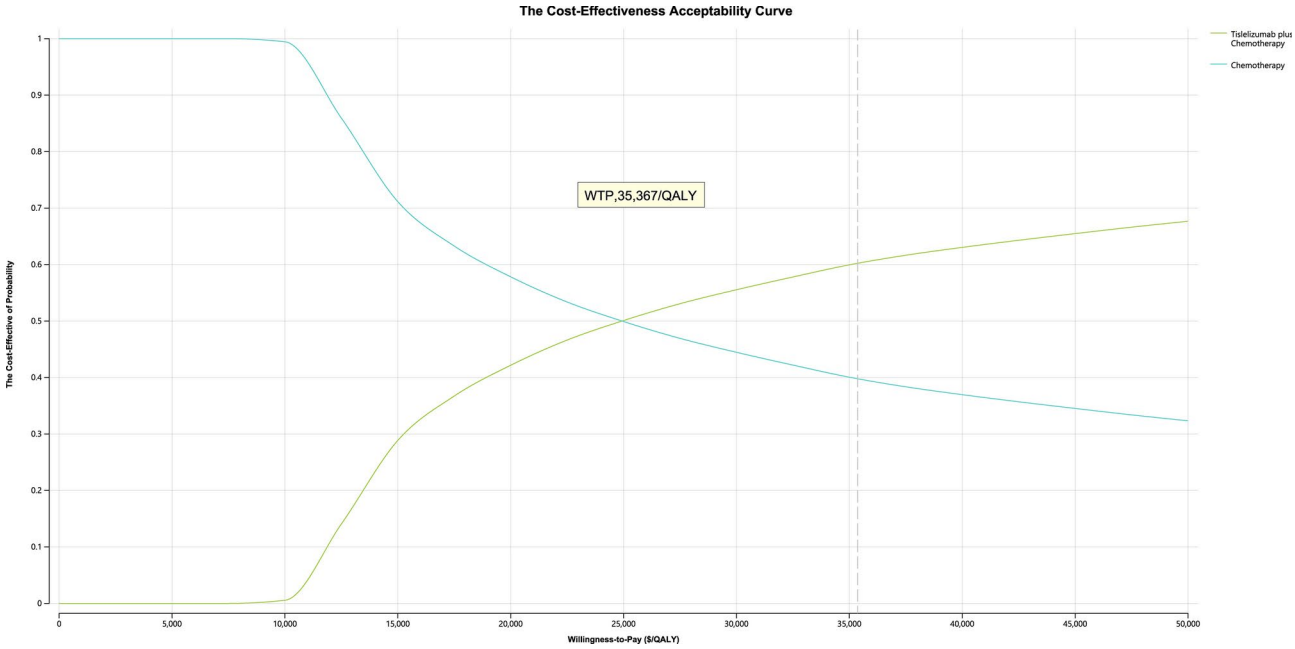

Fig. 2. The cost-effectiveness acceptability curves for tislelizumab plus chemotherapy strategy compared to chemotherapy strategy in Chinese patients. *WTP* willingness-to-pay, *QALY* quality-adjusted life-year.

Although significant progress has been made, there is still ample opportunity for additional investigation and improvement in immunotherapy for SCLC. First, the investigations IMpower133 and CASPIAN primarily consisted of individuals of Caucasian ancestry, with minimal inclusion of Asian populations. Second, there is still room for improvement in the effectiveness of immunotherapy in SCLC compared to non-SCLC. Therefore, it is necessary to develop therapeutic options that are both more efficient and safer to address the unique treatment requirements of patients with SCLC specifically. Furthermore, medical insurance does not cover imported immunotherapeutic medicines, making them comparatively expensive. Considering that the financial cost of treating patients with SCLC is significantly more than those with non-SCLC, the economic feasibility of treatment programs becomes a crucial factor for both doctors and patients[31].

Subgroup	OS HR (95% CI)	ICER, $/QALY	INHB, QALYs	Cost-effectiveness probability of tislelizumab at WTP, %
				$35,367/QALY (%)
Age group				
< 65 years	0.76 (0.58–0.99)	40,083	−0.02	43.3
≥ 65 years	0.74 (0.53–1.03)	37,859	−0.01	45.5
Sex				
Male	0.76 (0.61–0.96)	40,083	−0.02	43.3
Female	0.69 (0.42–1.12)	33,329	0.01	50.8
ECOG Performance status				
0	0.74 (0.42–1.29)	37,859	−0.01	45.5
1	0.76 (0.60–0.95)	40,083	−0.02	43.3
Smoking status				
Never	0.70 (0.45–1.08)	34,138	0.01	50.2
Smoker	0.77 (0.60–0.97)	41,308	−0.03	42.6
AJCC staging at study entry				
III	0.62(0.30–1.29)	28,643	0.06	55.5
IV	0.75 (0.60–0.93)	38,936	−0.02	44.4
Liver metastases at baseline				
Yes	0.65 (0.44–0.95)	30,469	0.04	53.9
No	0.76 (0.59–0.98)	40,083	−0.02	43.3
Brain metastasis				
No	0.76 (0.61–0.93)	40,083	−0.02	43.3
Baseline LDH				
≤ ULN	0.73 (0.54–0.99)	36,847	−0.003	47.3
> ULN	0.80 (0.60–1.06)	45,549	−0.04	37.6
Choice of platinum				
Cisplatin	0.81(0.51–1.26)	47,190	−0.05	35.9
Carboplatin	0.74(0.59–0.94)	37,859	−0.01	45.5

Table 3. Results of subgroup analyses. *CI* confidence interval, *OS HR* overall survival hazard ratio, *ICER* incremental cost-effectiveness ratio, *INHB* incremental net health benefits, *QALY* quality-adjusted life-year, *WTP* Willingness-to-pay, *ECOG* Eastern Cooperative Oncology Group, *AJCC* American Joint Committee on Cancer, *LDH* lactate dehydrogenase, *ULN* upper limit of normal.

The phase 3 RATIONALE 312 trial investigates the efficacy of tislelizumab, a PD-1 monoclonal antibody, in treating ES-SCLC in the Chinese population. The close connection to Chinese clinical practice and patient needs has undeniably brought a significant boost to investigating PD-1 inhibitors for treating ES-SCLC. This has resulted in new evidence supporting the use of PD-1 inhibitors in combination with chemotherapy, thereby expanding the range of treatment choices available for ES-SCLC patients. The success of the RATIONALE-312 expanded the group of patients who can benefit from tislelizumab in lung cancer, including both NSCLC and SCLC. Furthermore, tislelizumab is now recognized as the only immunotherapy medicine worldwide that offers comprehensive advantages for lung cancer. Therefore, we initially implemented a Markov model to compare the potential cost-effectiveness of chemotherapy with and without tislelizumab, considering the viewpoint of Chinese payers in the present economic environment.

The present study showed that selecting tislelizumab in conjunction with chemotherapy as the initial treatment for ES-SCLC patients was more economically efficient than choosing chemotherapy alone. The ICER was calculated to be $23,553 per QALY, significantly lower than China's WTP of $35,367 per QALY. These findings provide support for tislelizumab combined with chemotherapy as a more affordable therapy option. The findings of the one-way sensitivity analysis showed that the health utility value of the PD had the greatest influence on the ICER. Despite shifting within specific ranges, this parameter did not significantly change the model results, thereby confirming the reliability of our model. By setting the weighted total cost (WTP) for all Chinese patients at $25,000 per QALY, probabilistic sensitivity analysis demonstrated that tislelizumab combined with chemotherapy was more affordable than chemotherapy alone.

The WTP threshold is commonly utilized to assess the monetary value that healthcare customers are willing to allocate for healthcare benefits in cost-effectiveness assessments[32]. In China, there are significant differences in per capita GDP among different regions due to its large territory and uneven economic development. In 2022, the WTP thresholds for a QALY in Beijing, Shanghai, Zhejiang, Guangdong, Hubei, Hunan, Guizhou, Tibet, Heilongjiang, and Gansu were $78,451/QALY, $74,035/QALY, $49,023/QALY, $42,011/QALY, $38,049/QALY, $30,332/QALY, $21,583/QALY, $24,059/QALY, $21,006/QALY, and $18,571/QALY in 2022, respectively[33]. Notably, in most areas of China (29/34), the most economically efficient treatment option may be the combination

of tislelizumab and chemotherapy. Drug producers should develop comprehensive support programs that consider the variations in GDP value and local WTP levels across different provinces within a country.

Socioeconomic status (SES) also has a considerable impact on the results of cost-effectiveness analysis. Previous studies have established a link between lung cancer and SES, showing that populations in poverty have lower rates of screening, treatment, and survival[34]. A social welfare analysis by Aurelie Meunier et al.[35], which categorized lung cancer patients by SES, found that opportunity cost thresholds and patient population characteristics were major determinants of the overall distributional cost-effectiveness results, with SES significantly impacting the opportunity cost threshold. In our analysis, although the RATIONALE-312 trial did not categorize patients by SES, the WTP threshold would be lower for those in lower SES groups, thus reducing the opportunity for cost-effectiveness. However, it does not imply that patients are required to select less effective treatment options.

Immunotherapy drugs, known as ICIs, have shown enormous potential as effective treatments. As a result, healthcare policymakers, researchers, and physicians are increasingly interested in the economic aspects of different treatment alternatives. In 2022, Kang et al.[22] conducted a study to evaluate five economic immunotherapies (combined with chemotherapy) as initial treatments for ES-SCLC in China[28,29,36–38]. The final results demonstrated that atezolizumab, in combination with chemotherapy, proved to be the most economically advantageous option compared to other ICI combinations with chemotherapy[22]. This analysis offers various benefits compared to prior studies. First, based on the most recent clinical data, this study is the first to evaluate the affordability of tislelizumab and chemotherapy for ES-SCLC patients in China. Second, the RATIONALE 312 study exclusively involves Chinese participants, making it readily applicable to the general Chinese population and closer to the clinical practices and needs of Chinese patients. Third, the result of this study reflected the clinical landscape for ES-SCLC patients, which can help direct future strategic goals in the pursuit of a balance between improved patient health and cost savings—a crucial consideration for a nation struggling with limited healthcare resources.

This study also has some limitations. First, the proposed model is based exclusively on the most recent data from the RATIONALE 312 trial and this trial-based study struggles to accurately simulate natural disease progression in the real world and cannot fully reflect the efficacy and resource consumption observed in clinical practice. Hence, any biases present in this trial will be evident in the present research results. Second, even though the model was proven stable through sensitivity analysis, the long-term advantages of the two treatment regimens are inferred from existing survival curves. Meanwhile, additional real-world data remains crucial for predicting future economic outcomes. Third, due to the unavailability of complete QoL statistical data in the trial, our model primarily relies on health state utility values obtained from the published research. Single sensitivity variables that reflect utility factors frequently influence the stability of the model, perhaps leading to inflated or underestimated utility values. However, sensitivity analysis indicates that model outcomes remain unchanged when utility value parameters vary within defined ranges. Fourth, the analysis included only the most frequent TEAEs with grade 3–4. While adverse events associated with immunotherapy, such as encephalitis and myocarditis, are rare, their management can be very costly, potentially underestimating the direct medical costs. Therefore, including all AEs with an incidence greater than 1% in the study would provide a more accurate assessment of the costs associated with AEs management for patients. Finally, the model cannot be applied to all patient subgroups due to the absence of accessible survival curves, and the results of the subgroup analyses based on subgroup-specified HRs may differ significantly from actual survival outcomes, and thus, these findings should be interpreted with caution. Nevertheless, further follow-up analysis is necessary for these specific categories of patients.

Conclusion

The present study found that combining tislelizumab with chemotherapy was a more favorable treatment option for Chinese patients with ES-SCLC as a first-line treatment compared to using chemotherapy alone. This finding has essential implications for the Chinese healthcare system and clinical practice.

Data availability

All authors had full access to all of the data in this study and took complete responsibility for the integrity of the data and accuracy of the data analysis. The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Author contributions
L.R. and C.F.P. designed the experiment. L.R. and C.F.P. performed the experiments. L.R. and C.F.P. analyzed the data. C.F.P. contributed analysis tools and funding. L.R. and C.F.P. wrote the manuscript. All authors have read and approved the manuscript.

Declarations

Competing interests
The authors declare no competing interests.

Ethics approval and consent to participate

All the data included in this analysis were derived from published literature and public data. No patient-identifiable data were applied or used. Therefore, the approval of the independent ethics committee was not required.

Additional information

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Correspondence and requests for materials should be addressed to F.C.

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